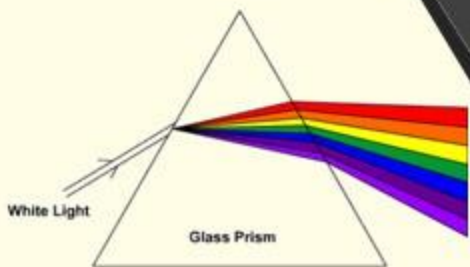


# PHYSICS

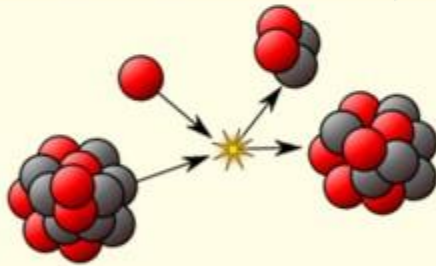
## GRAVITATION

### MODULE - 14

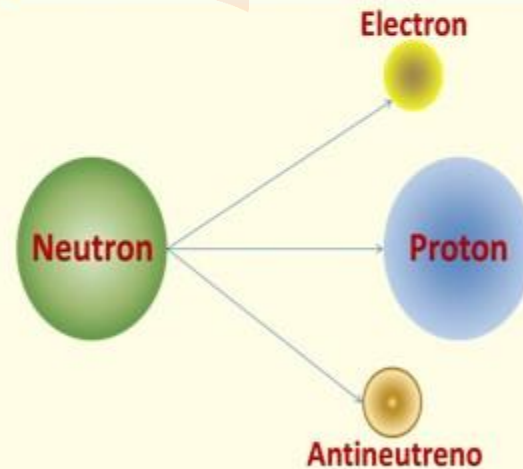


# FUNDAMENTAL FORCES OF NATURE

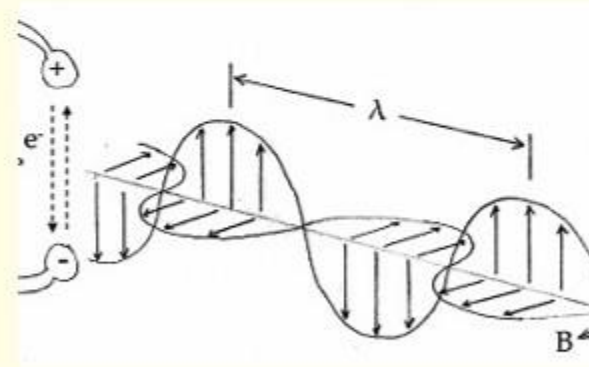
**STRONG  
NUCLEAR FORCE**



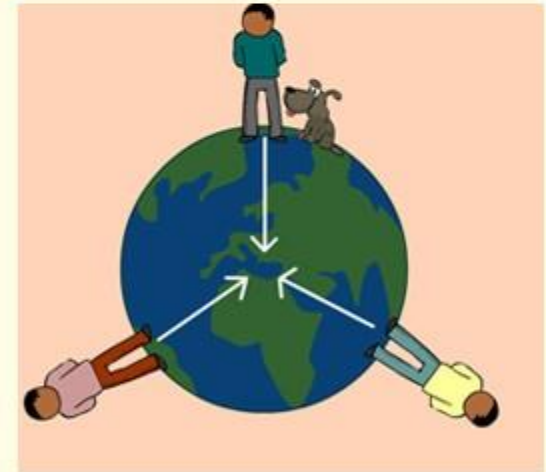
**WEAK  
NUCLEAR FORCE**



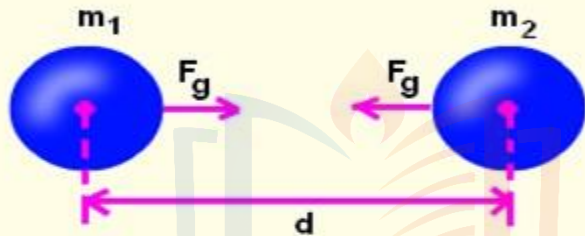
**ELECTROMAGNETIC  
FORCE**



**GRAVITATIONAL  
FORCE**



## GRAVITATION



$$F_g = \frac{Gm_1m_2}{r^2}$$

- Every object in the universe attracts every other object with a force which is proportional to the product of their masses.
- The Force is inversely proportional to the square value of the distance between the two masses

Gravitational constant  $G = 6.673 \times 10^{-11} \text{ Nm}^2 \text{ Kg}^{-2}$

*Gravitational force is weakest amongst all the forces*



## WHY MOON DOES NOT HIT EARTH DUE TO GRAVITATIONAL PULL?



- The motion of the moon around the Earth is due to the centripetal force.
- The Centripetal force or the centre seeking force is provided by the force of attraction of Earth.
- If there were no such force, The moon would pursue a uniform straight line of motion and flown off.



## CONTRASTING FEATURES

### CENTRIFUGAL FORCE

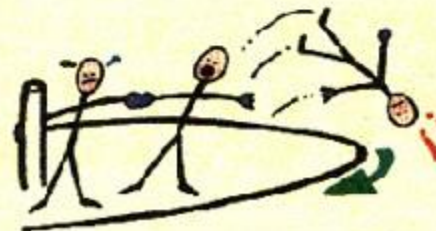
- Tendency of an object following a curved path to fly away because of centrifugal force.
- Centrifugal force is not a real force.
- It happens on account of inertia

### CENTRIPETAL FORCE

- The force that keeps an object moving with a uniform speed in a circular path.
- Centripetal force is a real force
- It is applied from the centre of a curved path of an object.



**CENTRIPETAL FORCE**



**CENTRIFUGAL FORCE**



## COMMON APPLICATIONS

### CENTRIFUGAL FORCE



When water is removed from clothes and clothes are dried in the machine, basic principle is centrifugal force acts on water.

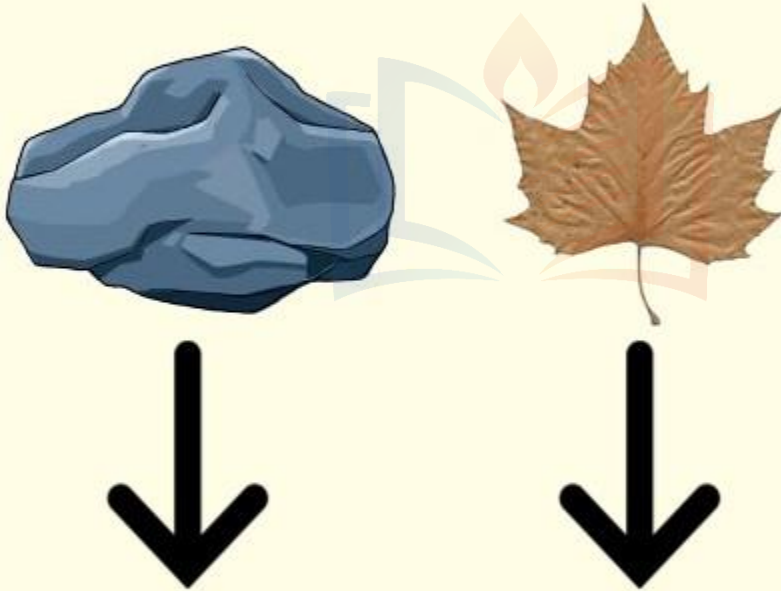
### CENTRIPETAL FORCE



*As a car makes a turn, the force of friction acting upon the turned wheels of the car provides centripetal force required for circular motion*

## THE CONCEPT OF FREE FALL

### WHO WILL FALL FAST IN IDEAL CONDITIONS?



- All the objects during free fall experiences acceleration due to gravity.
- The acceleration experienced is independent of its mass.
- All objects heavy or small fall at the same rate.

But what happens in real condition?



## THE CONCEPT OF FREE FALL

### WHAT WILL HAPPEN IN REAL CONDITIONS?



- In real life, there is air resistance.
- As falling objects speed up, the force of air resistance increases.
- When the air resistance gets as strong as the force of gravity, the falling objects stops accelerating.



**QUESTION 1**

A thief stole a box with valuable article of weight 'W' and jumped down a wall of height h. Before he reach the ground he experienced a load of

- (a) zero.
- (b)  $W / 2$
- (c) W
- (d)  $2 W$

**ANSWER :** In the condition of free fall apparent weight becomes Zero.



## CONCEPT OF MASS AND WEIGHT

### MASS

- Mass is the amount of matter in an object.
- Mass is a fundamental quantity
- Mass is equal at all places in Earth and in Space or in other planets
- Mass is not defined by the acceleration due to gravity



Weight on Earth - 560 N



MASS IS 56 KG

### WEIGHT

- Weight is the force of gravity which acts upon mass.
- Weight is a derived quantity
- Weight varies at different points on Earth, It is zero in space.
- Weight is defined by the acceleration due to gravity



Weight on Moon - 90 N



**QUESTION 2**

**Astronauts on the orbiting space station are weightless because...**

1. There is no gravity in space and they do not weigh anything.
2. Space is a vacuum and there is no gravity in a vacuum.
3. Space is a vacuum and there is no air resistance in a vacuum.
4. The astronauts are far from Earth's surface at a location where gravitation has a minimal effect.

***Which of the above are correct options:***

- a) 1, 2 and 3
- b) 1, 3 and 4
- c) 1, 2, 3 and 4.
- d) 1 and 3 only



## IN WHICH SITUATION THE DENT IS DEEPER?



**PLACING THE FOOT ON  
SAND**



**LAYING THE WHOLE  
BODY ON SAND**

*The dent is deeper in the 1<sup>st</sup> case where the entire weight of the body came to the sand from a smaller surface area.*

## THRUST AND PRESSURE



- The force acting on an object perpendicular to the surface is called thrust.
- The effect of thrust is larger when it is acted on smaller surface area.
- The effect of thrust is smaller when it is acted on a larger surface area.
- Thus Pressure is defined as the thrust acting on per unit area.

*Thrust applied by a elephant's foot is huge compared to a ladies sandal, but because of the tiny surface area of the sandal, the pressure applied by a lady on the road is much higher than the elephant's foot!!!*

## APPLICATIONS OF PRESSURE



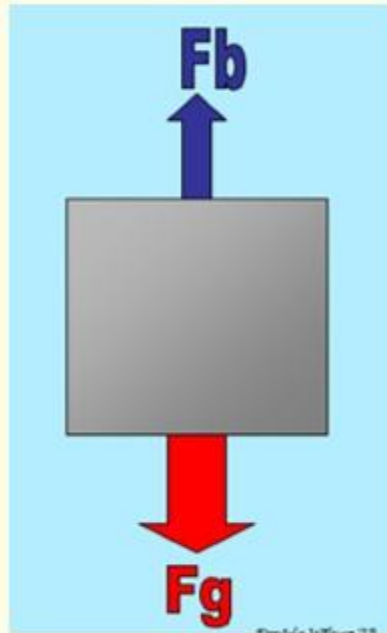
**1** A sharp knife can cut easily than a blunt one because of very less surface area of the polished edge.

**2** A nail can penetrate very easily while hammering against a wall than wood because of sharp ends.

**3** A ski makes it possible for the skier to stay on the surface of the snow, thus enabling the skier to glide more effectively down a snow-covered hill than walking in shoes.



## BUOYANT FORCE



- When we push a body in water, we experience an upward force which is exerted by the water.
- This upward force exerted by any fluid is called buoyant force.
- The magnitude of the buoyant force depends on
  - The density of the fluid
  - The volume of the displaced fluid.

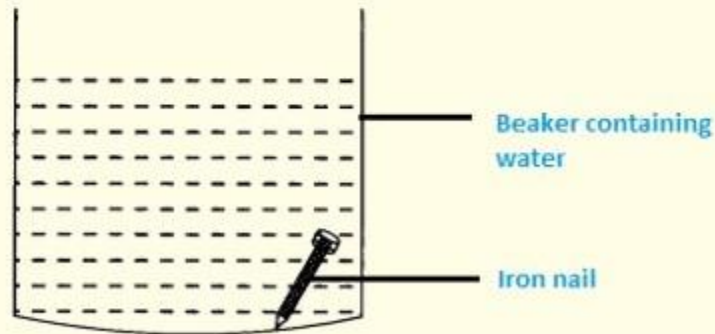


## WHY WE CAN FLOAT IN THE DEAD SEA?



- The buoyant force increases as the density of the fluid increases.
- The dead sea water is denser than regular water or salt water.
- This provides more buoyant force by the fluid on the immersed body.

## IF AN IRON SINKS, HOW CAN A LARGE SHIP STAY AFLOAT?

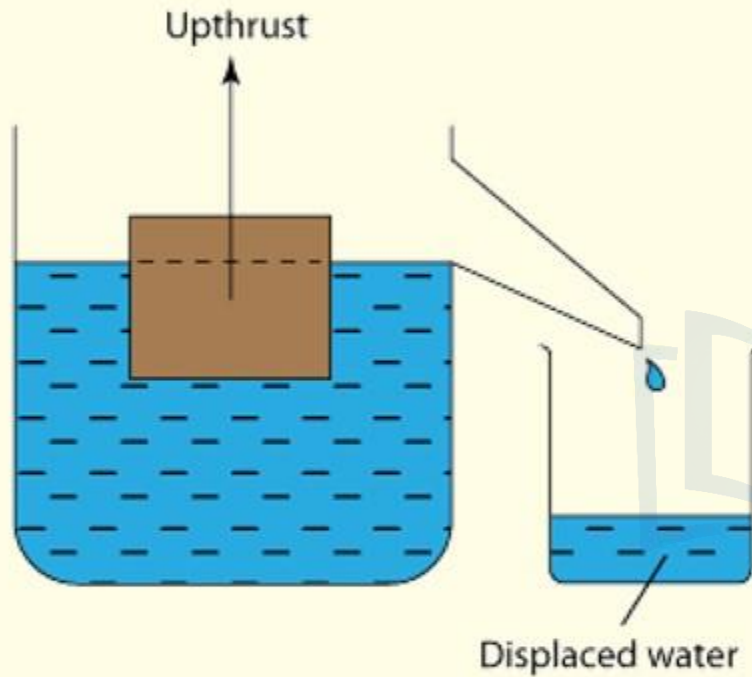


- This attributes to the volume of the displaced fluid.
- A nail can displace a very small volume of water, thus, it sinks.
- A ship displaces huge volume of water, than its mass, thus it can float.
- Anything that floats in water will have lesser density than water.

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$



## ARCHIMEDES PRINCIPLE

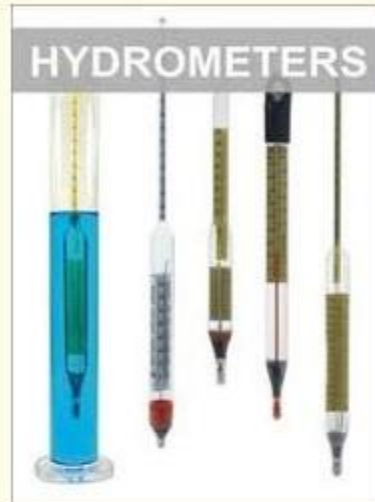


*The upthrust is equal to the water displaced.*

- Archimedes principle relates buoyancy to displacement.
- Simply put, the weight of the displaced liquid is the weight of the object.
- The weight of the displaced liquid is the buoyant force.



## ARCHIMEDES PRINCIPLE - APPLICATIONS



- 1** Archimedes principle is used in hydrometers. Hydrometers are used to determine the purity of liquids.
- 2** The principle is also used in Lactometers, which are particularly used to measure the purity of milk.
- 3** Archimedes principle is also used in selecting appropriate construction material for bodies which are meant to float as ships and boats.



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